

Sensyte

Project Proposal

Water Treatment Plant – PLC Data Monitoring & Motor Control System

Client: TechSync Solutions (Pvt) Ltd

Elevating Your Digital Presence

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Who we are

Sensyte (Pvt) Limited is a Sri Lanka based technology company formed through the merger of CircuitUp Technologies & Innvotics (pvt) Limited, delivering end-to-end engineering and product solutions across electronics, IoT, renewable energy, and electric vehicle (EV) industries.

We specialize in turnkey product development, covering electronic R&D, PCB and schematic design, embedded firmware and software development, IoT platforms, web dashboards, and OEM services. Beyond core electronics, Sensyte provides complete system solutions for renewable energy and EV-based products, from concept to deployment.

With strong expertise in multi-layer PCB design (up to 12 layers), power and high-speed electronics, ultra-low power devices, wireless communication, industrial systems, and energy and EV solutions, our multidisciplinary team delivers reliable, scalable, and future-ready technologies.

Sensyte serves clients across Australia, New Zealand, Singapore, the USA, and Sri Lanka, partnering closely to transform ideas into impactful, sustainable solutions.

Sensyte partners with innovators worldwide to turn complex ideas into reliable, production-ready solutions, delivering engineering excellence with global reach and real-world impact.

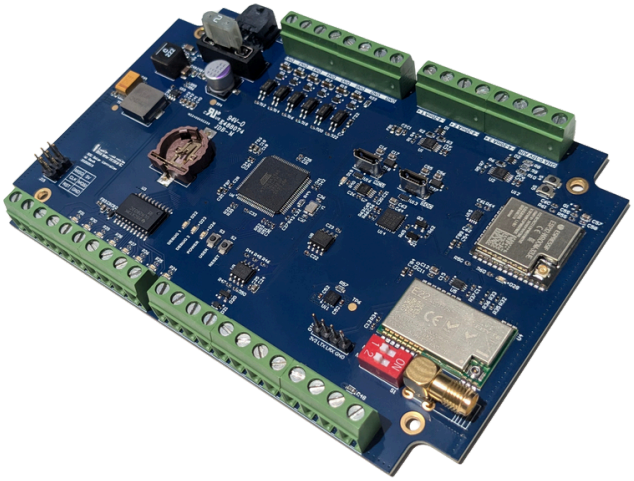
100+
Projects Completed

50+
Customers

NZ AUS SG USA
Overseas Projects

Our Services

Sensyte has experienced engineering to facilitate a variety of customer requirements. Including PCB design services, Firmware Development, Standalone software, Web Application, and Manufacturing services.



Electronic Product Development

Sensyte delivers complete electronic product development from concept and system design to hardware, firmware, and enclosure development. We design multilayer PCBs, develop reliable embedded firmware, and create production-ready enclosures, taking products from prototype to mass production. Our solutions support IoT, energy, wireless, and industrial applications, with a strong focus on performance, reliability, and manufacturability.



Mechanical Design

Sensyte offers comprehensive mechanical design and enclosure development services, including concept design, surface modeling, structural and thermal simulations, and design optimization for manufacturing. We support rapid prototyping through in-house 3D printing and coordinate injection molding for mass production. Using industry-leading tools such as SolidWorks, our team ensures mechanically robust, functional, and aesthetically refined designs ready for real-world deployment.

Our Services

Software Development

Sensyte provides full-stack software development to complement and scale electronic products. We develop secure and high-performance web dashboards, desktop applications, and mobile apps for both Android and iOS—enabling seamless device management, data visualization, and user interaction.

Our web solutions are built using modern technologies such as PHP, JavaScript, and Python, with frameworks like Laravel and Node.js, and integrated with cloud services, APIs, and SQL/NoSQL databases. We also develop Windows-based desktop applications using C# for device configuration, control, and specialized client workflows. From real-time monitoring to analytics-driven platforms, our software is designed to be reliable, scalable, and production-ready.



Proposed Solution

Overview

Sensyte proposes the design, development and deployment of an IoT-based PLC data acquisition and remote motor control system for the Client's Water Treatment Plant (WTP) network, covering five (5) locations.

At each location, a Sensyte industrial gateway connects to the existing Programmable Logic Controller (PLC) and continuously reads the selected process data points (flow, level, pressure, water quality, pump and alarm status). The same gateway provides two (2) relay outputs that allow authorised operators to start and stop two (2) pump motors remotely.

Data from all five locations is sent securely to a central cloud SCADA platform, giving the operations team one live view of the whole plant network from a web dashboard or mobile app — without replacing the existing PLCs or their control logic.

Objectives of the System

By deploying Sensyte's PLC gateway devices, the Client can expect the following services and capabilities:

Real-Time Process Monitoring

- Live reading of PLC data points from all five (5) locations
- Raw and treated water flow rates and totalised volumes
- Tank, sump and reservoir levels and line pressures
- Water quality: pH, turbidity, residual chlorine, conductivity
- Pump and motor run / stop / trip status, run hours and energy
- Historical data logging, trend analysis and daily reports

Proposed Solution

Objectives of the System cntd.

Remote Motor Control

- Remote START / STOP of two (2) motors per location via relay outputs
- Run-feedback confirmation for every command
- Automatic control based on tank level set-points and schedules
- Duty / standby pump rotation to balance run hours
- Respect for local Hand / Off / Auto selectors and PLC interlocks

Alarms & Notifications

- Low / high level, dry-run, overload, phase-failure and trip alarms
- Water quality limit alarms (turbidity, pH, chlorine)
- PLC communication loss and power failure alerts
- Instant alerts via SMS, e-mail and mobile push notification

Site Coverage – Five (5) Locations

One Wi-Fi gateway is installed in the PLC / control panel at each location, controlling two (2) motors. The data point count is the proposed baseline and will be finalised after the site survey.

Location	Gateway	PLC Data Points	Relay Outputs	Feedback Inputs
Location 01	1	Up to 64	2	2
Location 02	1	Up to 64	2	2
Location 03	1	Up to 64	2	2
Location 04	1	Up to 64	2	2
Location 05	1	Up to 64	2	2
Total	**5**	**Up to 320**	**10**	**10**

Proposed Solution

PLC Data Points

The gateway polls the PLC over Modbus RTU (RS-485) or Modbus TCP (Ethernet). Other PLC protocols can be supported depending on the PLC make and model. Typical data points for a water treatment plant are listed below.

Category	Typical Data Points	Unit / Type
Flow	Raw water inflow, treated water outflow, totalised flow	m^3/h, m^3
Level	Raw water sump, clarifier, filter, clear-water reservoir, elevated tank	m, %
Pressure	Pump discharge, distribution header, filter differential pressure	bar
Water Quality	pH, turbidity, residual chlorine, conductivity / TDS	pH, NTU, mg/L, µS/cm
Pumps & Motors	Run / stop / trip status, Auto / Manual mode, run hours, start count	Digital, h
Electrical	Motor current, voltage, power, energy, power factor	A, V, kW, kWh
Chemical Dosing	Dosing pump status, chemical tank level, dosing rate	Digital, %, L/h
Valves & Filters	Valve open / close position, backwash status, blower status	Digital
Alarms	Low / high level, dry-run, overload, phase failure, high turbidity, low chlorine	Digital
Panel & Power	Mains power status, UPS / battery, panel temperature, PLC health	Digital, °C

The final tag list for each location will be prepared after reviewing the PLC program or Modbus register map. Data points are read without changing the existing PLC logic. Only designated command registers / relay outputs are used for control.

Proposed Solution

Motor Control Methods

Two control methods are offered. The most suitable method will be selected for each motor after the site survey.

A – Relay Output Control

Gateway relay contacts are wired into the motor starter / MCC remote start-stop circuit. Suitable where the PLC program cannot be modified.

B – PLC Command Control

Start / stop commands are written to designated PLC registers, and the PLC runs the motor using its existing interlocks. Recommended where the PLC can be updated.

Safety & Control Philosophy

- Local Hand / Off / Auto selector always has priority over remote commands
- Overload, dry-run and phase-failure protections stay hard-wired in circuit
- Momentary (pulse) commands with run-feedback check and timeout alarm
- Configurable fail-safe state on network loss or gateway restart
- Two-step confirmation, role-based permissions and full command audit log

SCADA & Centralized Data Management

- Network overview of all five (5) locations with live status
- Plant mimic screens with pump control buttons per location
- Historical trends, daily production and energy reports (PDF / Excel)
- Alarm and event management with acknowledgement
- User role management, API access and cloud hosting on AWS

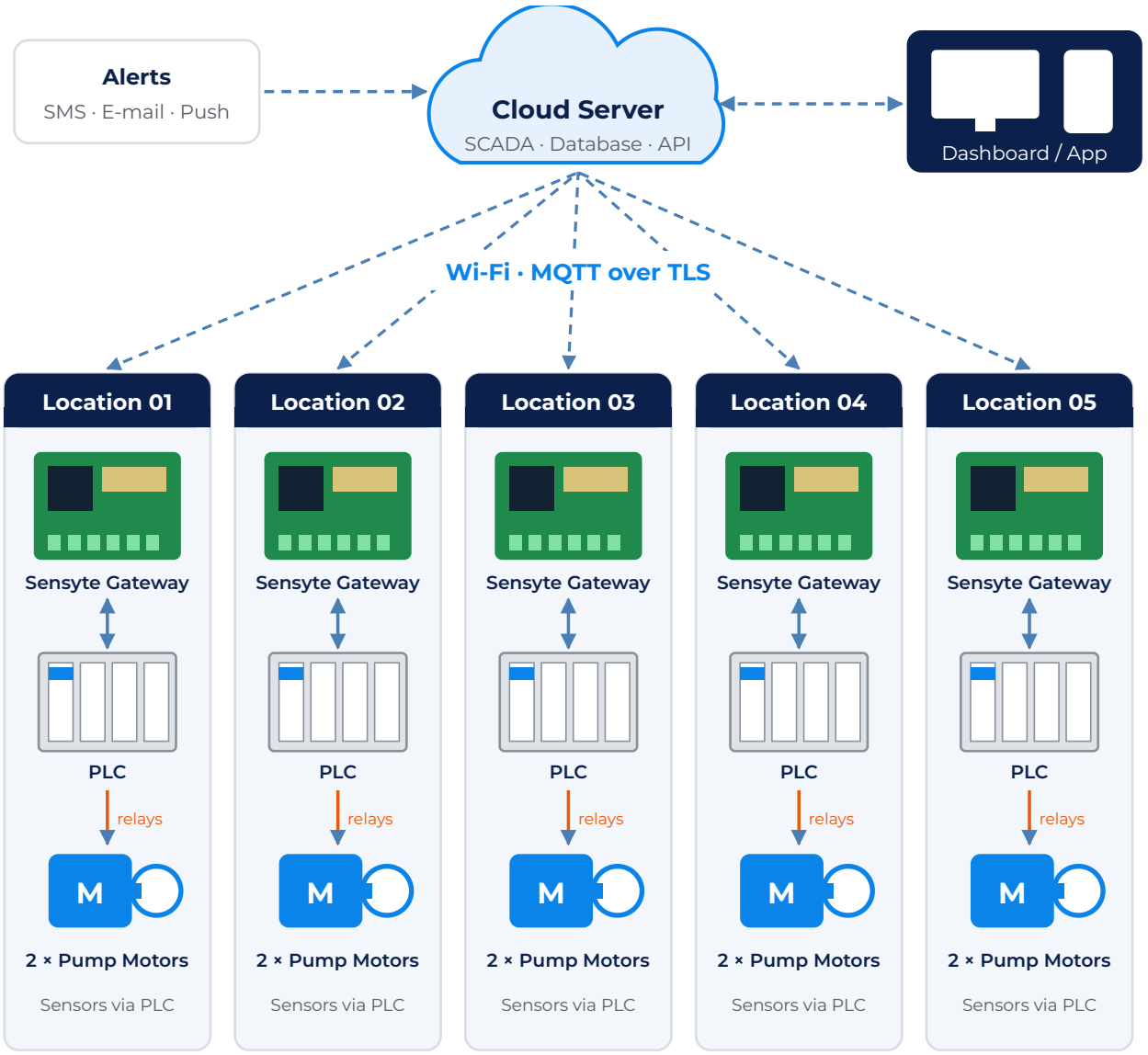
Value Proposition

- Works with the existing PLCs – no replacement of plant control systems
- One platform for all five (5) locations
- Faster response to faults and fewer site visits
- Lower energy and chemical costs through data-driven operation

System Architecture

Overall Architecture – Five (5) Locations

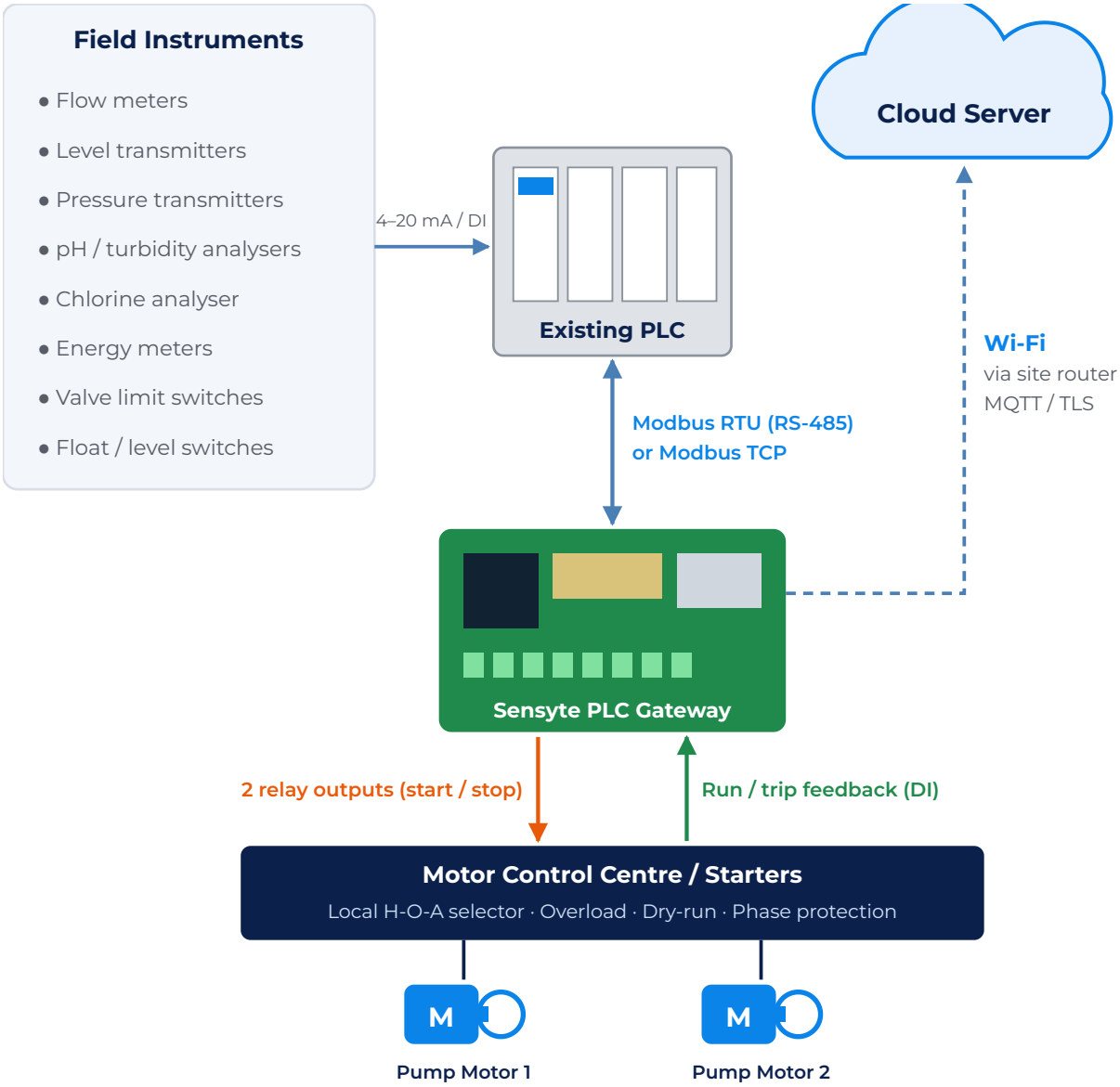
Each location has a Sensyte Wi-Fi gateway connected to the local PLC and to two (2) motor control circuits. The gateways connect to the site Wi-Fi network and send data to the cloud server using encrypted MQTT. Operators monitor and control all sites from the web dashboard or mobile app.



System Architecture

Single Location – Detailed View

Field instruments remain connected to the existing PLC. The gateway reads PLC data over Modbus and drives motor starters through its relay outputs, while auxiliary contacts return run / trip feedback to the gateway's digital inputs.



System Architecture

Gateway Specification (proposed)

Feature	Specification
PLC Interface	1× RS-485 (Modbus RTU master), 1× Ethernet (Modbus TCP client)
Relay Outputs	2× dry-contact relays (one per motor), 5 A @ 250 VAC, with LED indication
Digital Inputs	2× opto-isolated inputs for motor run / trip feedback
Analog Inputs	2× 4–20 mA (optional, for independent sensors)
Connectivity	Wi-Fi 2.4 GHz (IEEE 802.11 b/g/n) to the site router
Data Security	MQTT over TLS, device authentication
Data Buffering	Local storage with store-and-forward during outages
Configuration	BLE mobile app set-up, remote (OTA) firmware updates
Power & Mounting	24 VDC, surge protected, DIN-rail mount in control panel

Implementation Plan

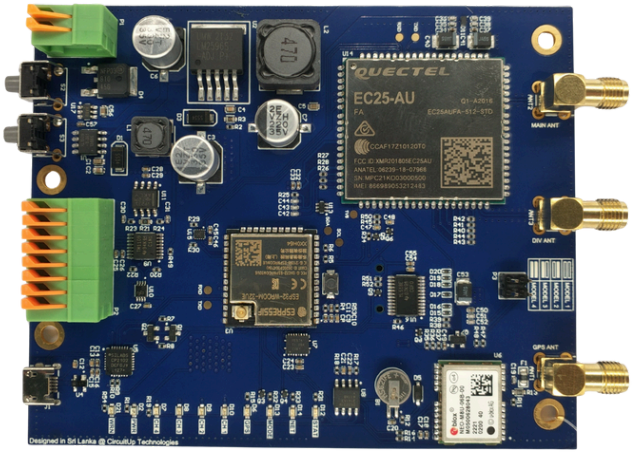
#	Phase	Timeline
1	Site survey and PLC register mapping (5 locations)	Weeks 1 – 2
2	Gateway hardware build and firmware development	Weeks 2 – 6
3	SCADA dashboard and mobile app development	Weeks 3 – 9
4	Factory acceptance test with PLC simulator	Weeks 8 – 9
5	Installation and commissioning at all locations	Weeks 9 – 11
6	Site acceptance test, training and handover	Week 12

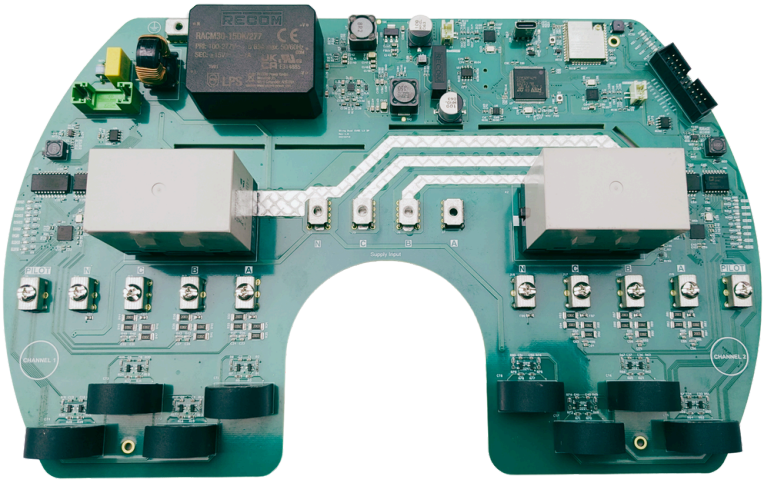
Project Portfolio

Hardware

Remote EV Battery Diagnostic Tool

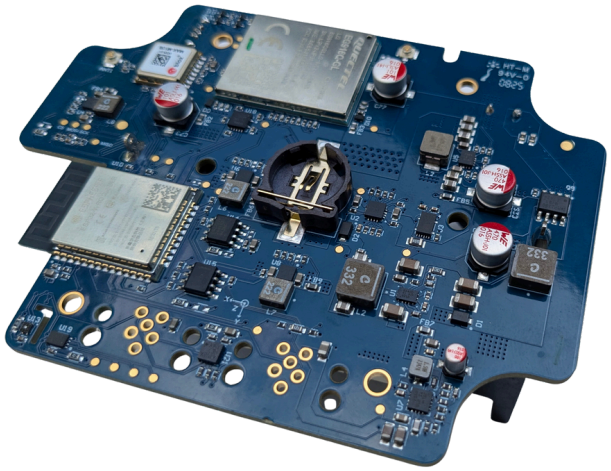
This project was designed as a remote EV battery health monitoring and diagnostic tool. CAN bus takes battery information and stores it in an SD card. Whenever the user requests, the logged file will be uploaded to the server using 4G.





EV Charger

An innovative dual-port Level 2 charger designed with the advanced power monitoring features and the wireless connectivity.

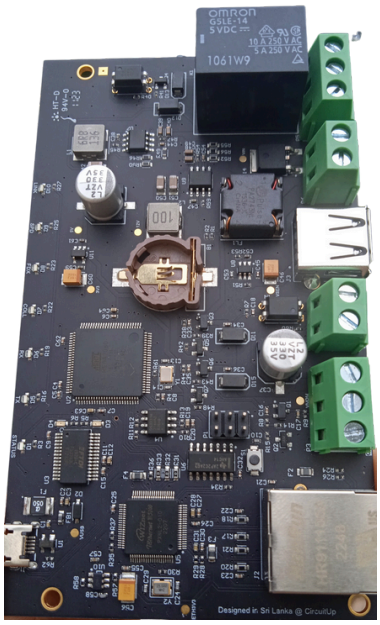


ULP Temperature Node

An ultra-low-power temperature device designed to operate for over one year on a single battery. It supports multiple sensors, FUOTA, and periodically sends data to the cloud over a 4G LTE network for reliable remote monitoring.

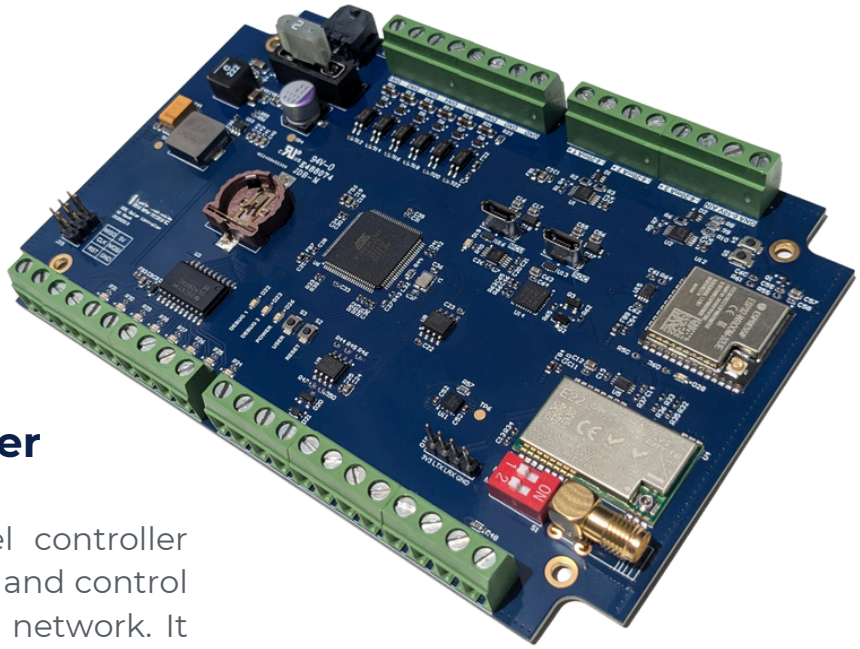
Project Portfolio

Hardware



Ethernet Sensor Controller

Designed to control specific sensors over the Ethernet. Also, multiple DC/DC convertors and power switches included.



Solar Panel Controller

An IoT-enabled solar panel controller enables remote monitoring and control of solar panels over a LoRa network. It supports multiple analog and digital inputs and outputs for robust operation, while built-in safety features ensure reliable and secure performance in industrial environments.

Project Portfolio

Mechanical



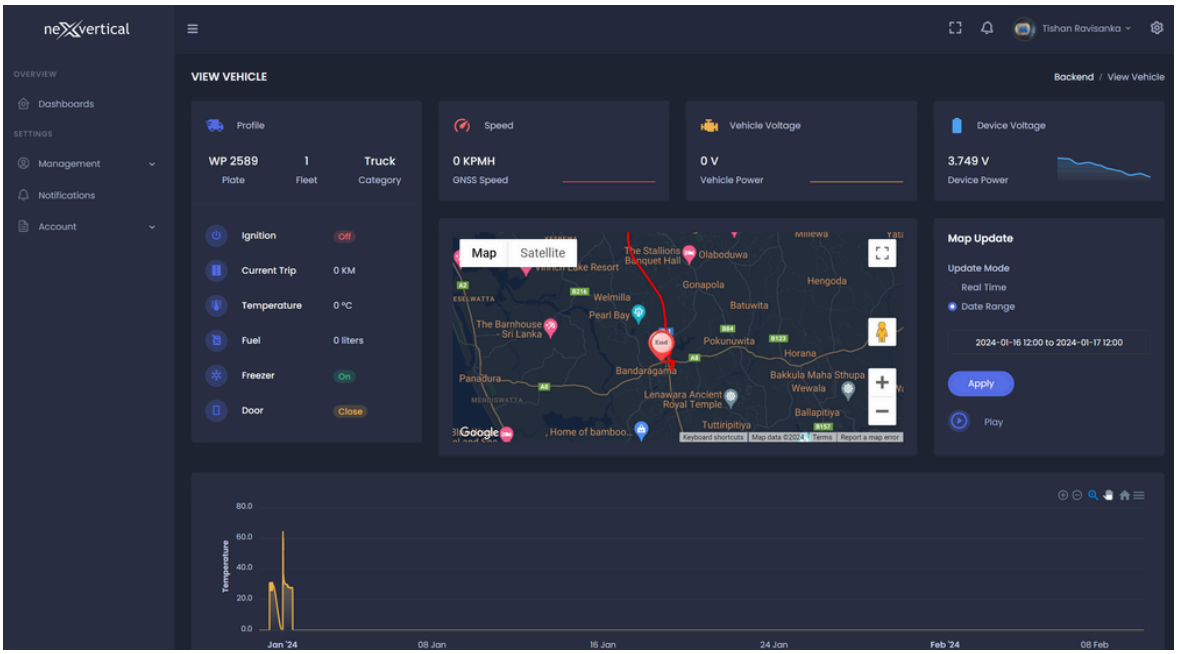
Enclosure Designs for LoRa WAN Temperature Node

Exclusively designed for a custom LoRaWAN PCB, this enclosure provides protection against dust and water splashes while offering two dedicated inlets for external connectors.

Project Portfolio

Software

Fleet Management System

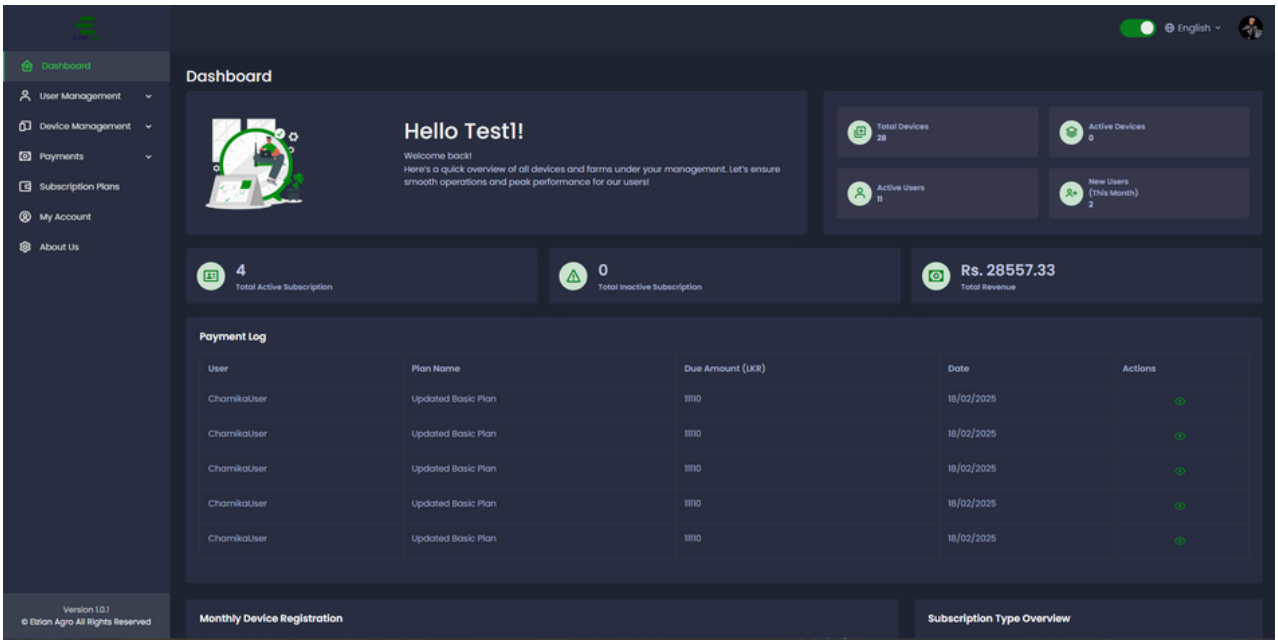


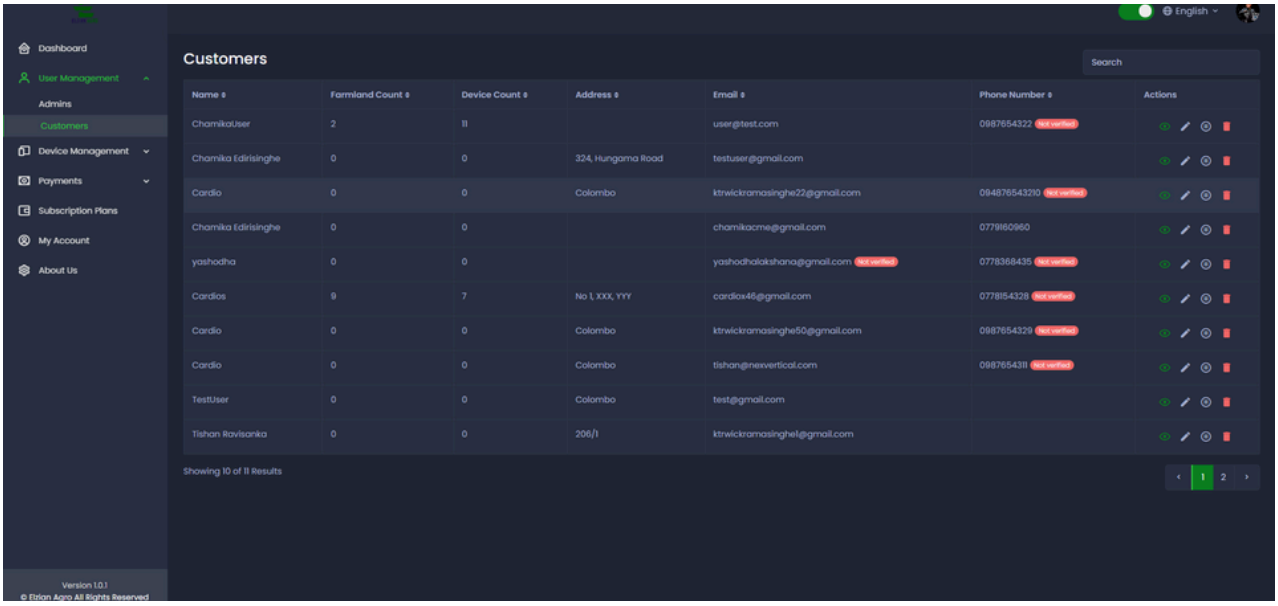


Project Portfolio

Software

IoT Platform for Smart Agriculture

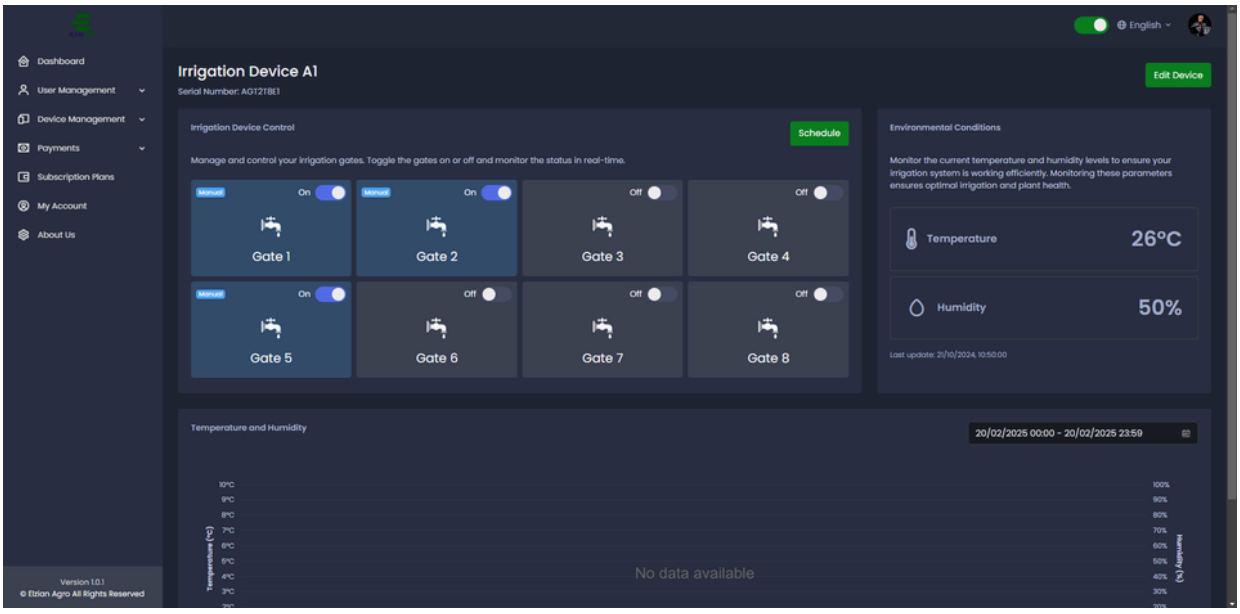


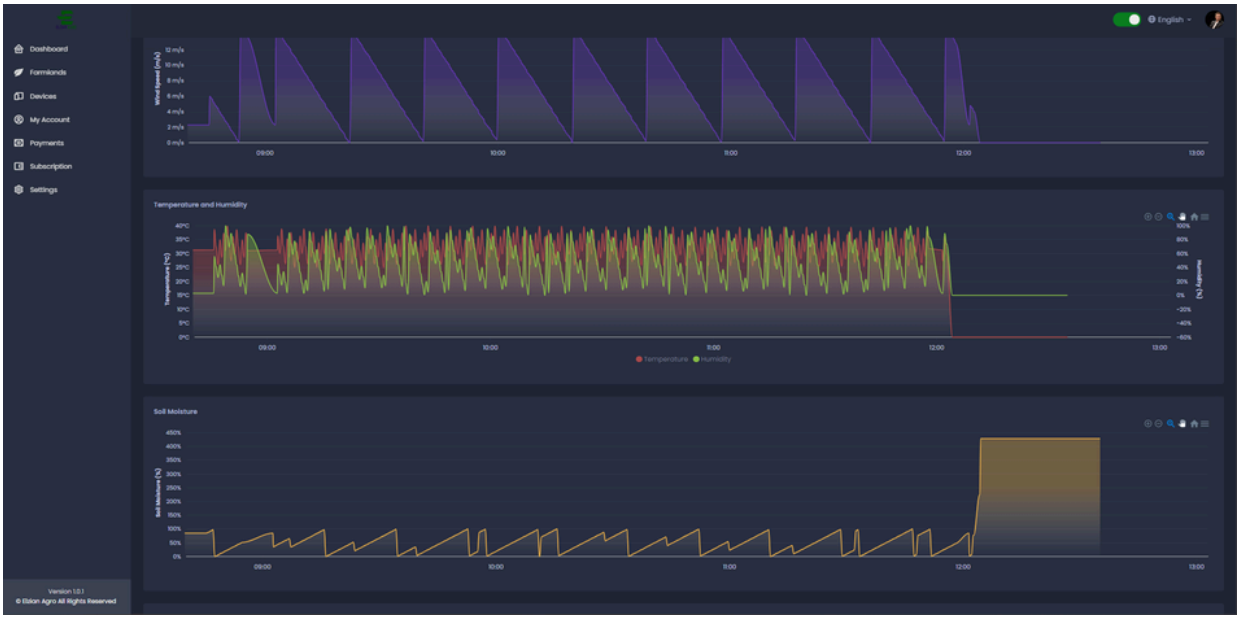


Project Portfolio

Software

IoT Platform for Smart Agriculture

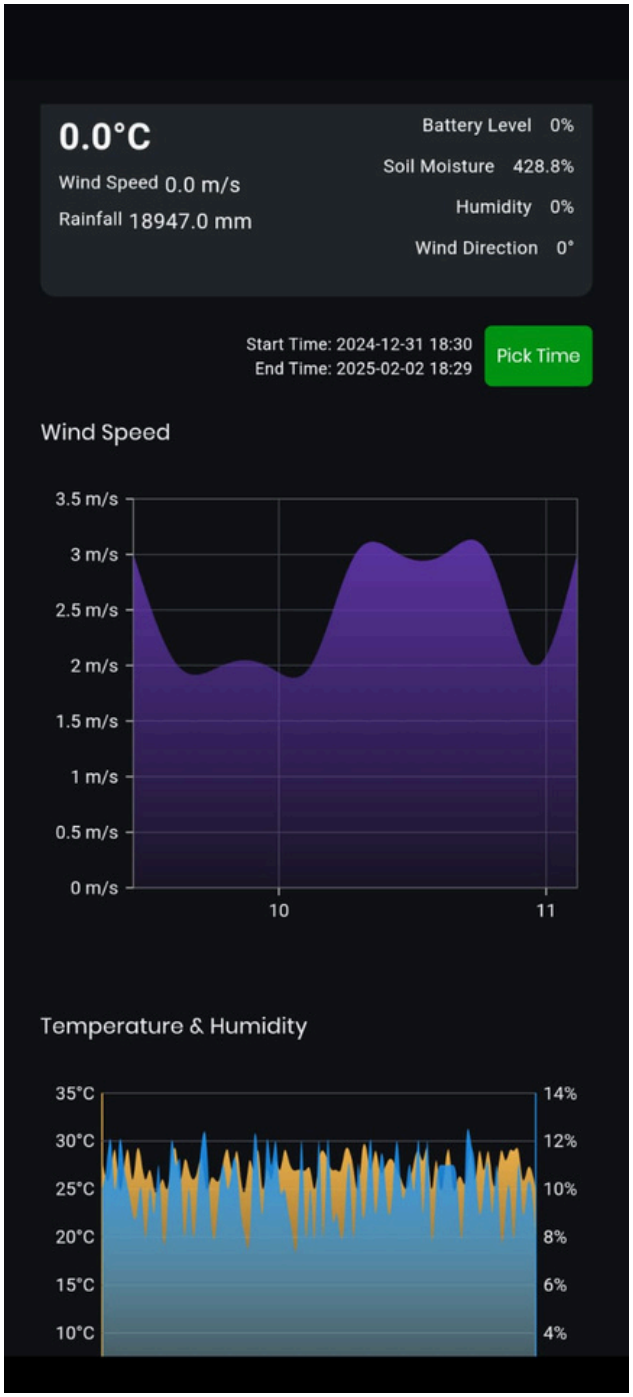


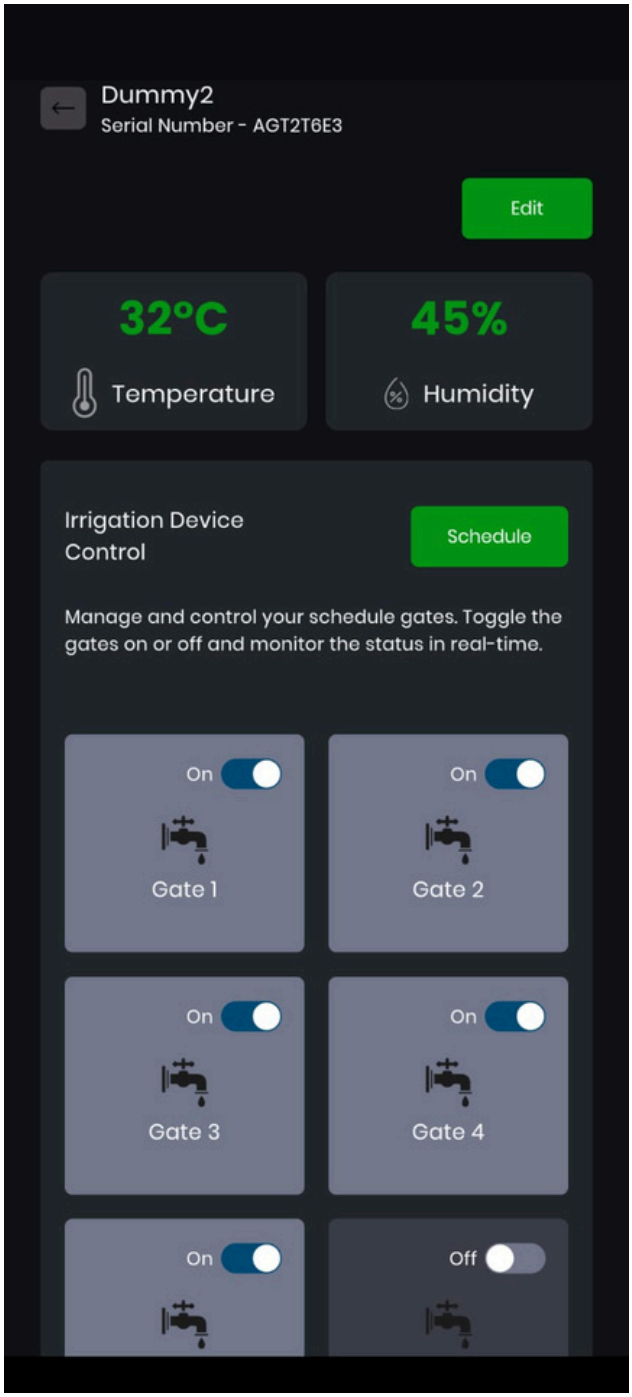


Project Portfolio

Software

IoT Platform for Smart Agriculture - Mobile App

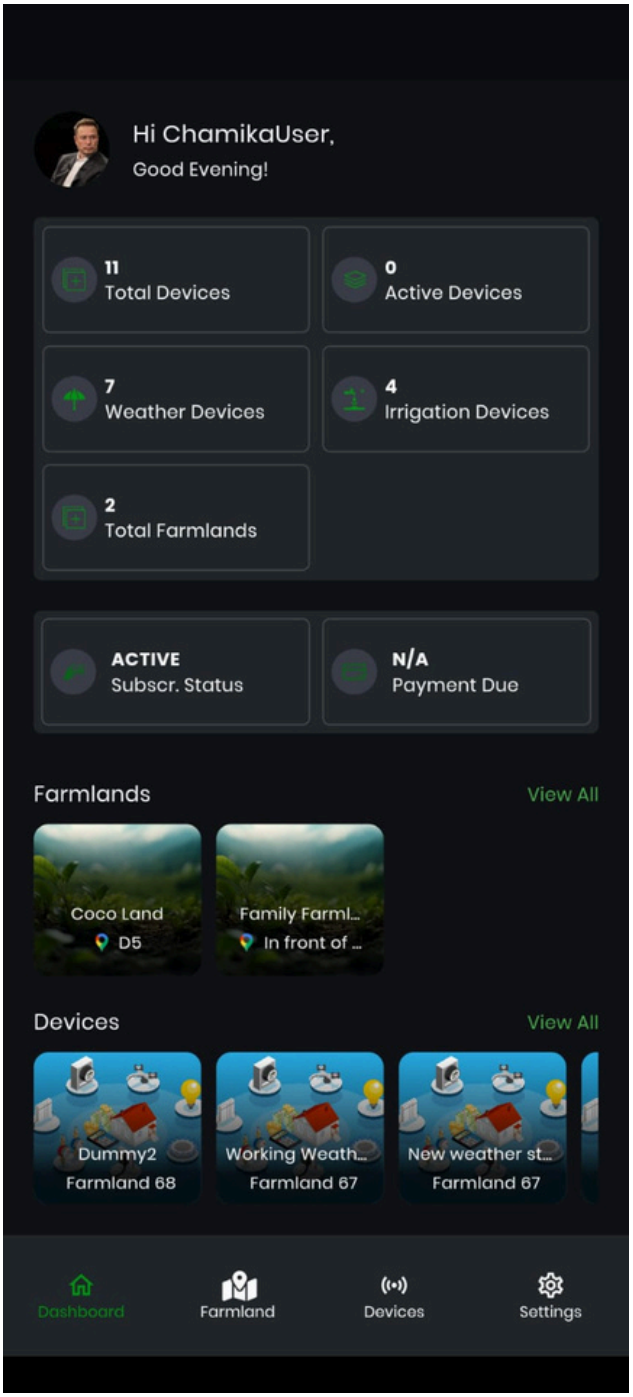


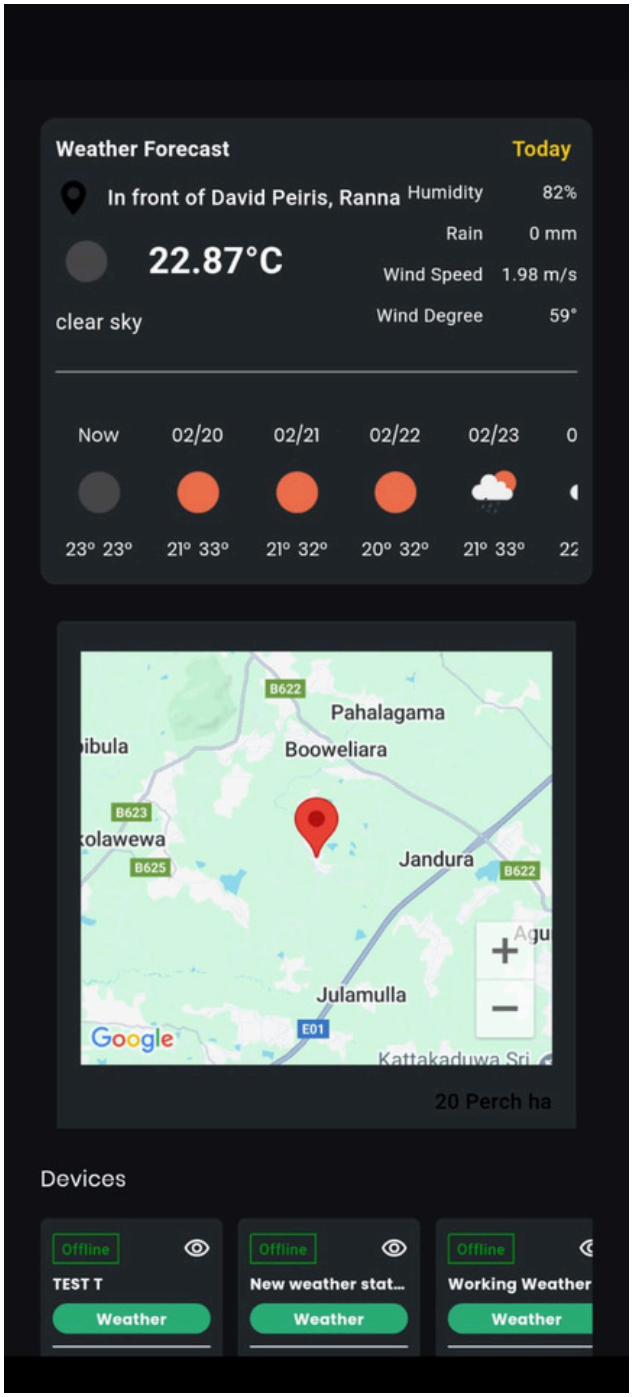


Project Portfolio

Software

IoT Platform for Smart Agriculture - Mobile App





Payment & Conditions

General Conditions

- The estimated project duration shall be up to three (3) months from the advance payment and receipt of the PLC information.
- A detailed and final project timeline shall be provided after the final technical and commercial discussion and mutual agreement.
- The scope of deliverables shall include the following:
  - **Gateway Devices** – Supply of five (5) Sensyte PLC gateway devices, one (1) per location, each with two (2) relay outputs to control two (2) motors, including configuration and commissioning.
  - **PLC Integration** – Reading of the agreed data points from the PLC at each location. The Client shall provide the PLC make / model, program or Modbus register map, and access to the panels. Changes to the existing PLC program shall be done by the Client or PLC vendor, or quoted separately.
  - **Mobile Application** – Monitoring, control and device configuration app delivered in APK (Android) format.
  - **Backend & Dashboard** – Deployment on an AWS server, including one (1) year of free hosting.
  - **Source Code & Project Files** – Not included in this agreement. May be made available upon additional payment, under a separate written agreement.
- Additional devices or locations may be supplied upon the Client's request, at prices confirmed in writing.
- Electrical wiring of relay outputs to motor starters / MCC panels shall be done by the Client's licensed electrician under Sensyte's guidance, or under a separate contract.
- A stable Wi-Fi network with internet access at each location shall be provided by the Client.

Payment & Conditions

Payment Conditions

- The Client shall pay 50% of the total project fee as an advance prior to the commencement of any work.
- The balance 50% of the total project fee shall be settled within four (4) weeks from the date on which the project deliverables are formally handed over to the Client.

warranty

- Warranty Period
  - A six (6) month warranty shall be provided from the date of sale.
- Warranty Coverage
  - The warranty covers manufacturing defects only.
  - The warranty does not cover physical damage or damage resulting from external factors, including but not limited to:
    - High-voltage spikes or electrical surges
    - Lightning or thunder strikes
    - Fire, water, or environmental damage
    - Improper use, mishandling, or unauthorized modifications
- Dashboard Bug Fixes
  - Any software bugs identified within six (6) months from the date of deployment in the dashboard system shall be fixed free of charge.
  - This applies only to defects related to the originally delivered functionality and scope.
- Dashboard Maintenance
  - Ongoing maintenance, updates, and support for the dashboard system may be provided under a separate annual maintenance contract, subject to mutually agreed terms and pricing.

Contact Us

Website
www.senstye.io

E-mail
yashodha@sentye.io

Phone
+94 77 836 8435

Social Media

HQ address
Sensyte PVT LTD
276/B Negombo Road, Thudella,
Ja-ela. 11350.
Sri Lanka

